

Dinakara Siva Srinivas Pathakota

• dinakar.pathakota@gmail.com • +91 9963817633 • Hyderabad, India • github.com/dinakar0745
• linkedin.com/in/dinakar-pathakota-32a823251/ • dp.ntechx.dev

RESEARCH INTERESTS

Computational pathology and whole-slide image (WSI) analysis; deep learning for histopathology and medical imaging; large-scale image pipelines and GPU-accelerated ML systems; graph neural networks. Long-term interest in building clinically useful, deployable models for digital pathology.

EDUCATION

Koneru Lakshmaiah Education Foundation (KL University) 2022 – 2026

B.Tech (Honors), Computer Science & Engineering — Specialization in Cyber Security & Blockchain · Hyderabad, India

- First Class with Distinction — CGPA **8.86 / 10**; 202.5 credits earned. Graduated April 2026.
- Relevant coursework: *machine learning, data structures & algorithms, computer vision / image processing, distributed systems, cryptography & security, blockchain.*

RESEARCH EXPERIENCE

Independent Research — Computational Pathology 2026 – Present

Self-directed research program building toward doctoral work in medical-imaging ML

- Building a WSI inference platform that runs detection models over gigapixel slides — tumour-cell and mitotic-figure detection, covering slide ingestion, tiling, model execution, and region-level result aggregation.
- Running a deep-learning study on lymph-node metastasis detection using the CAMELYON whole-slide dataset, targeting a preprint.
- Published a sole-author technical report asking whether robustness loss in pathology foundation models comes from the frozen encoder or the linear probe: evaluated Phikon, Phikon-v2, Kaiko ViT-B/16 and DINOv2-L under blur, JPEG compression and stain shift on NCT-CRC-HE and PatchCamelyon. A probe trained on mixed clean and degraded embeddings recovered most of the lost accuracy (Phikon-v2 at JPEG quality 15: 76.3% to 92.5% balanced accuracy) and changed which encoder ranks as most robust; code and full results released openly.
- Authored a technical and market brief on vertical WSI/ML for *veterinary and toxicologic (preclinical, non-regulated) pathology* — a domain with gigapixel imaging demands comparable to clinical pathology but far lower regulatory friction; currently moving from scoping into implementation.

NTechX Founder

Applied research in AI/ML and security

- Developed a graph neural network (GNN) model for automated smart-contract vulnerability auditing; contributed the core detection module for an academic manuscript.
- Led the venture end-to-end: research direction, model development, and technical execution.

MANUSCRIPTS & RESEARCH OUTPUT

Is It the Encoder or the Probe? Acquisition-Artifact Robustness of Pathology Foundation Models. Technical report, Zenodo, doi:10.5281/zenodo.23162508 *Published, Oct 2026*

Deep-learning detection of lymph-node metastases on whole-slide images (CAMELYON) *Ongoing*

PROFESSIONAL EXPERIENCE

Systems Group Jun 2026 – Sep 2026

Software Development Engineer — Hyderabad, India

- Built scalable internal applications and backend systems; shipped production services with FastAPI, PostgreSQL, and Docker.
- Built in-house software solutions for the parent company, Saridena Constructions, and delivered customized tools and applications tailored to client requirements.

Engineering Intern — Whole-Slide Imaging (WSI) Pipelines

- Engineered DICOM-based ingestion and processing pipelines for gigapixel whole-slide histopathology images, supporting both sparse and fully-tiled acquisition modes.
- Built GPU assignment and scheduling scripts to distribute tile-processing workloads across devices, improving throughput on large slide volumes.
- Designed RabbitMQ message routing across Python microservices to coordinate acquisition, tiling, and downstream image-processing stages.
- Contributed C++ acquisition modules interfacing with microscopy hardware, integrated into the end-to-end imaging pipeline.

SELECTED PROJECTS

WSI detection platform

- Platform for running AI detection models across whole-slide images, pathology model execution over gigapixel inputs with tiled inference and result aggregation.

Agricultural crop-image platform — PJTSAU*Professor Jayashankar Telangana State Agricultural University · KL University*

- Built a mobile application enabling farmers to upload crop images for expert review by university faculty; designed the system to curate a labeled agricultural image dataset for training predictive AI models toward automated crop diagnosis and support.

Smart-farming weed detection*KL University*

- Computer-vision weed-detection system on an IoT field platform, classifying crop versus weed from field imagery.

NEXUS OS — bootable AI operating environment

- Scaffolded a ReAct-style agent orchestrator with skill auto-discovery, a permission model, and a FastAPI server packaged as a bootable Linux distribution; extended with vision, retrieval-augmented generation (RAG), scheduling, and homelab monitoring.

Local document-AI extraction pipeline

- Built an on-premises pipeline (pdfplumber + Tesseract OCR + local LLM via Ollama) for structured extraction from scanned documents, with canonical item matching and a comparison dashboard — running fully locally on a GPU workstation.

Distributed inference cluster

- Two-node heterogeneous inference cluster (macOS + Windows/WSL) using exo, exploring model sharding and peer discovery for local large-model serving across commodity hardware.

Autonomous security-testing agent deployment

- Deployed and configured an autonomous penetration-testing agent against a locally-hosted open-weight LLM backend, keeping the full loop offline.

“orion” research homelab

- Self-administered Linux workstation with RAID storage, zero-trust remote access, and GPU-backed local LLM inference — the compute substrate for the imaging, document-AI, and agent work above.

TECHNICAL SKILLS

ML & Medical Imaging: Deep learning, pathology foundation models, linear probing and robustness evaluation, graph neural networks, retrieval-augmented generation, agent orchestration; whole-slide / gigapixel image processing, DICOM, image tiling, GPU compute, local & distributed LLM inference.

Languages: Python, C++, Java, JavaScript / TypeScript, SQL.

Systems & Infrastructure: Docker, FastAPI, NestJS, PostgreSQL, MongoDB, RabbitMQ, Firebase, Linux / systemd, RAID storage, GitHub Actions CI, VM deployment.

Cloud: AWS, GCP, Azure.

HONORS & AWARDS

- First Class with Distinction, B.Tech CSE (CGPA 8.86 / 10).
- Two-time hackathon winner.

CERTIFICATIONS

- AWS Certified Cloud Practitioner (CLF-C02)
- Microsoft Certified: Azure Fundamentals
- Automation Anywhere Certified Advanced RPA Professional
- GitHub Foundations
- Google Associate Cloud Engineer

LEADERSHIP & COMMUNITY

- Head, Cybersecurity Club — KL University.
- Founder, NTechX — AI/ML and cybersecurity venture.
- Founder / organizer, 00:00 (Zero Hundred Hours) — youth entrepreneurship community.